Mikołaj Lubiak

C++ Systems Engineer

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Profiles

GitHub
LinkedIn

Open Source

Intel Cryptography Primitives Library
Contributed a performance and correctness fix to Intel's cryptography-primitives (PR #95): made SHA-256 hash-method initialization thread-friendly by avoiding redundant updates to a shared function-pointer structure, added an early-return optimization for the sha256_ni path, updated changelog/version and formatting, and validated successful DPC++ compilation.

Result: microbenchmarks show ~10% CPU time reduction, ~32% fewer cache references and ~50% fewer instructions in targeted multi-threaded hashing workloads.

FTXUI
Implemented SliderOption::on_change in FTXUI (PR #938), adding a callback for slider value changes to improve component interactivity and usability.

Awards

Article in PagedOut! Magazine
Published article in Gynvael Coldwind's security and reverse-engineering zine.

Giganci Programowania
Applied Programming Contest, Faculty of Mathematics and Information Sciences — Warsaw University of Technology (age 17). Projects developed in C++

Skills

C/C++
Assembly
Python
Go
Embedded Linux (Yocto)
Bootloaders (U-Boot, custom x86 bootloader)
RTOS (FreeRTOS)
Microcontroller programming (ESP32)
Serial communication (SPI/I2C/UART)
Kernel programming
Filesystem programming (FAT12)
Vulkan
Real-time rendering
Multithreading
High performance computing
OpenCV
FFMPEG
GoogleTest
Git

Summary

C++ systems engineer focused on high-performance software, embedded systems and real-time graphics. Experience building low-latency applications, embedded Linux systems and microcontroller firmware. Strong interest in engine architecture, multithreading, rendering pipelines and low-level performance optimization. Recent work includes security provisioning and embedded Linux development for smart TV platforms, alongside personal projects including a Vulkan rendering engine, experimental operating system and ESP32 game console platform.

Experience

Xperi — Warsaw
Senior C++ Engineer — Security and Provisioning — 11.2025 - Present

- Developing low-latency user-space SDK and applications for smart TV platforms built on a custom Chromium fork.
- Implementing device security infrastructure and key provisioning mechanisms for embedded devices.
- Maintaining and extending a Yocto-based embedded Linux distribution used across device platforms.
- Hardening and extending a custom Das U-Boot fork used in the device boot chain.
- Working across the bootloader, Linux userspace and security infrastructure layers of the platform.

NielsenIQ — Warsaw
Mid Python Engineer — Data Collection Platform — 07.2025 - 11.2025

- Built large-scale web data collection pipelines for structured retail and product data.
- Developed high-reliability scraping infrastructure handling anti-bot protections and dynamic websites.
- Implemented data extraction, normalization and processing pipelines.

Lesson4Future
Backend Engineer (Intern → Lead Backend Engineer) — 08.2023 - 02.2024
Promoted from Backend Intern to Lead Backend Engineer; owned backend architecture, DevOps/sysadmin duties, and technical leadership for an educational platform delivered to the National Bank of Poland.

- Designed and implemented backend services and REST APIs (core derived from my open-source project Motey).
- Built CI/CD, automated deployments, monitoring, and incident runbooks; managed production infrastructure and security hardening.
- Ran daily standups, performed code reviews across the repo (backend & frontend), onboarded new hires and mentored engineers.
- Grew backend team and coordinated with frontend and pentest teams to deliver the platform on schedule.
- Drove all architecture decisions, set coding standards and ADRs, and ensured production quality until handoff.

Sealcode
Python Developer — Backend & Tooling — 07.2023 - 07.2025

- Developed backend services and internal tooling supporting company products.
- Built automation and data processing systems for internal workflows.
- Maintained and extended production backend components.

Go2Ops
Sysadmin / DevOps Engineer — 09.2023 - 10.2024

- Managed Linux infrastructure and deployments.
- Automated operational workflows and system administration tasks.
- Maintained monitoring, reliability and deployment pipelines.

Freelance
Software Developer — 2020 - 2025

- Delivered custom software solutions and automation tools.
- Worked across backend development, tooling and systems programming tasks.

Projects

Vulkan Game Engine
Real-time rendering engine written in C++ using Vulkan.

- Implemented rendering architecture with swapchains, pipelines and descriptor management.
- Built abstractions for GPU resources, frame synchronization and rendering pipeline.
- Implemented camera system, scene objects and real-time rendering loop.
- Focus on explicit GPU resource management and performance-oriented design.

ESPCon — ESP32 Game Console
Embedded game console platform for ESP32 microcontrollers.

- Designed lightweight CPU-based rendering engine for TFT displays.
- Implemented joystick and button input handling.
- RTOS-based architecture using FreeRTOS.
- Hardware communication through SPI.

GOS — Experimental x86 Operating System
Minimal operating system with custom bootloader and FAT12 filesystem.

- Implemented two-stage x86 real-mode bootloader using NASM and C.
- Developed minimal kernel with FAT12 filesystem support.
- Implemented disk access and filesystem parsing.
- Low-level work with BIOS interrupts and memory layout.

PixelPin — BLE E-Paper Display System

- Embedded firmware receiving custom protocol commands via Bluetooth Low Energy.
- Displays images on e-paper display driven by ESP32.
- Implemented state-machine architecture and custom transmission protocol.
- Optimized image dithering and bit-level operations for performance.

Terminal Animation Engine

- Multithreaded program converting video, GIF and images into animated ASCII rendering.
- Media processing using OpenCV and FFmpeg.
- Focus on parallel processing and performance optimization.

MathBoard — Handwriting Math Solver

- Platform that detects handwritten mathematical expressions and computationally solves them.
- OCR pipeline using EMNIST dataset, OpenCV, Tesseract and SymEngine.
- Cross-platform architecture with Unix sockets.